

Data, evidence and solutions for emerging disease



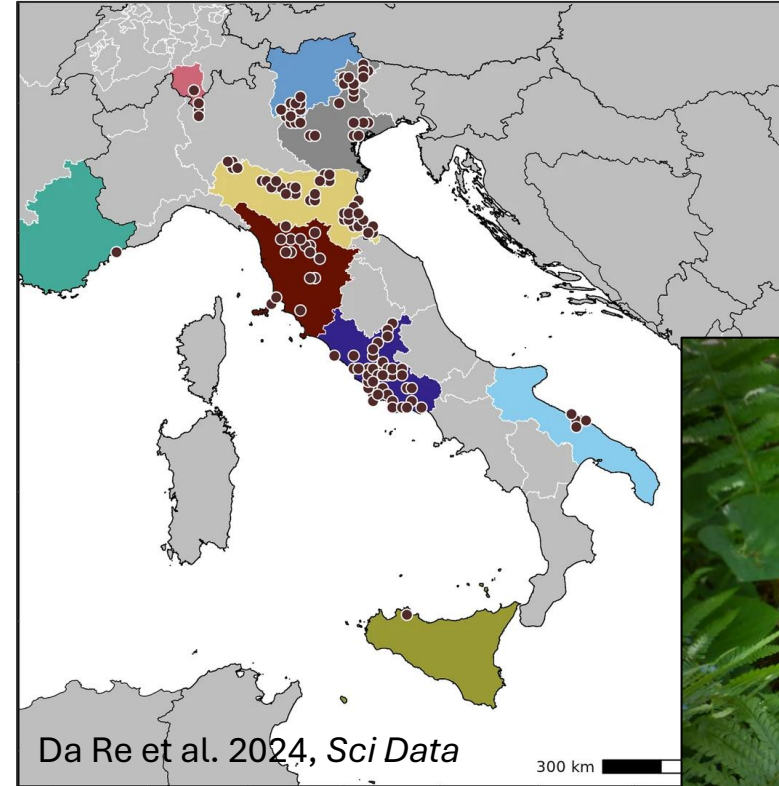
Emerging mosquito-borne diseases in Italy

Mosquito-borne diseases are surging in Europe – how worried are scientists?

Diseases such as West Nile virus and dengue are becoming increasingly common as the insects that spread them move north.

Italy has three chikungunya fever cases in Anzio

8 September 2017



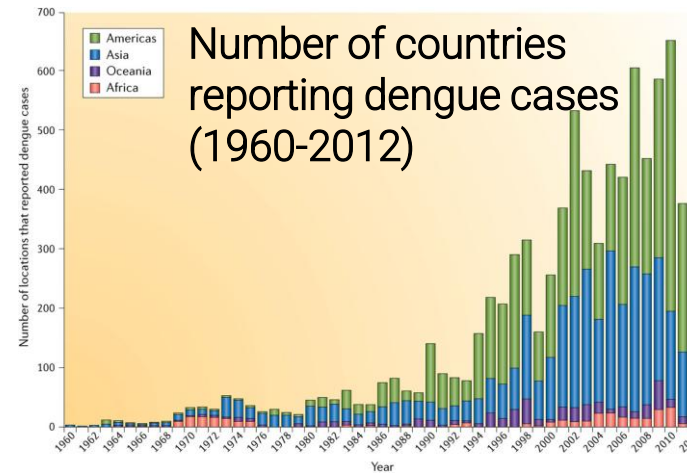
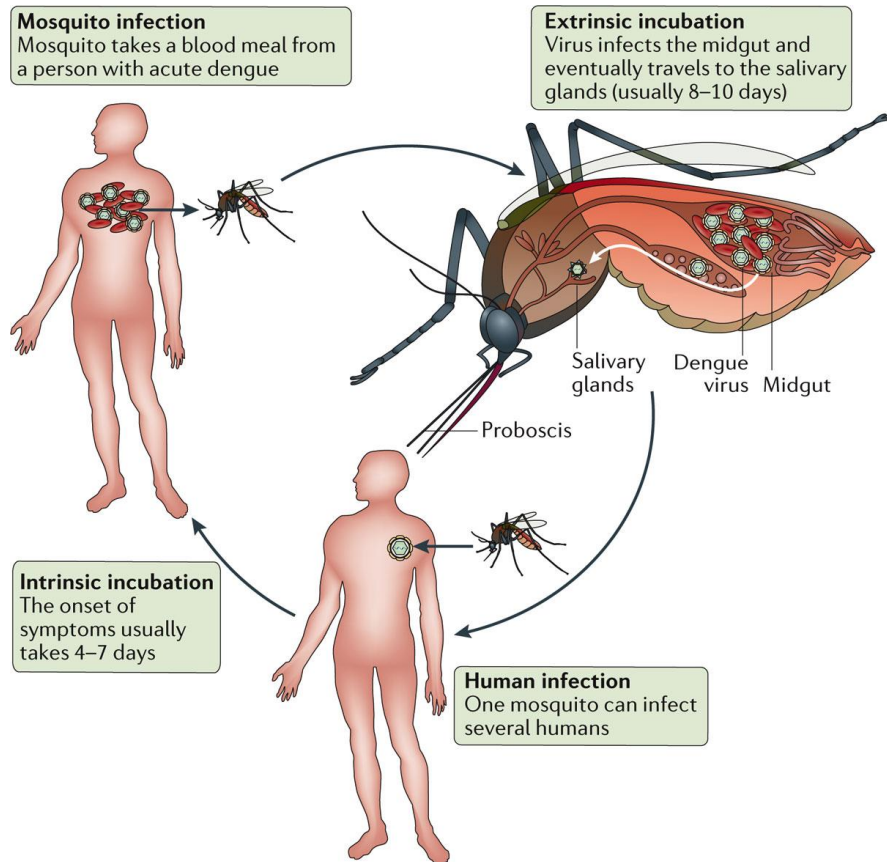
Should you worry about tiger mosquitos in Europe?



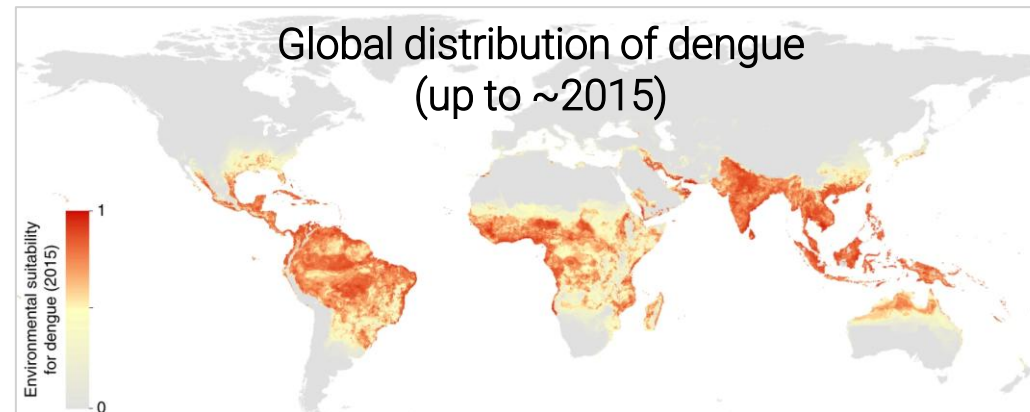
Carrying dengue fever and other diseases, the insects have arrived in Europe from Asia thanks to rising temperatures—here's what travelers should know.

Dengue: a one-slide primer

- Dengue virus (DENV) - mosquito-borne flavivirus with rapidly rising global burden
- Principal vector *Aedes aegypti*, but also *Ae. Albopictus*



Guzman et al. 2016, Nat Prim Dis



Messina et al 2019. Nat Rev Micro

Bangladesh dengue deaths cross 1,000 in worst outbreak on record

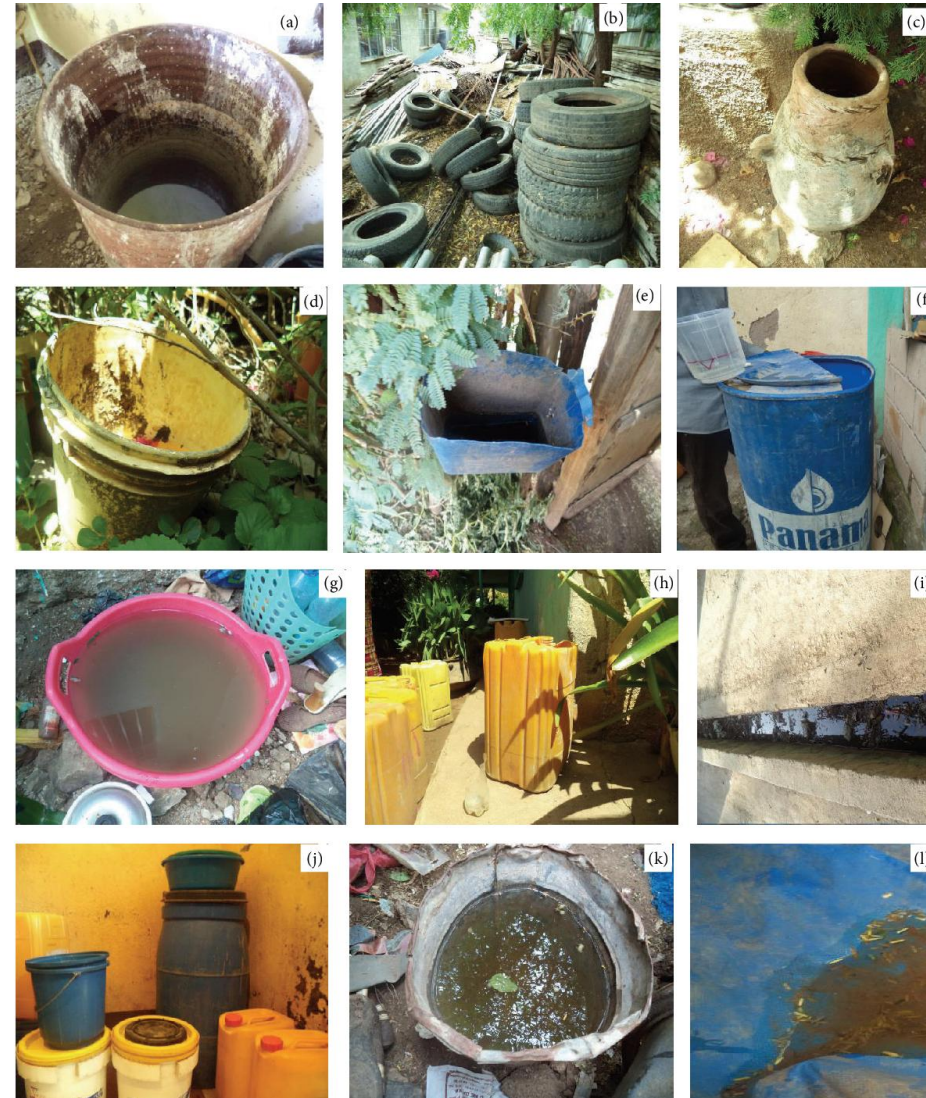
At least 1,017 die and nearly 209,000 are infected, making 2023 the deadliest year since the first recorded epidemic in 2000.



Aedes aegypti and albopictus



- Widespread and “synanthropic” (live near humans)
- Preferential human feeders (usually)
- Bite during day or at dusk/dawn (so mosquito nets don’t really help)
- Breed in human-made water containers of all kinds



The emergence of dengue in Italy

Cluster of 10 locally-transmitted cases near Rome (Central)

Vita et al 2024, Trop Med Inf Dis

Imported cases from overseas;
no local transmission

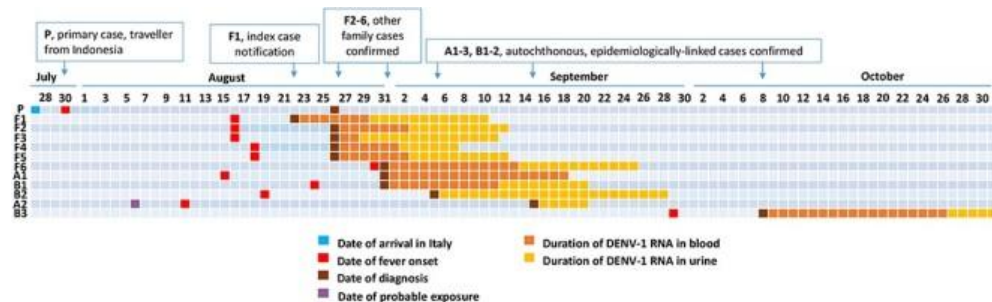
2010

2015

2020

2025

Outbreak of 11 locally-transmitted cases in Veneto (North)



Largest sustained local outbreak in Marche region (Central)

Barzon et al. 2021, Journal of Travel Medicine

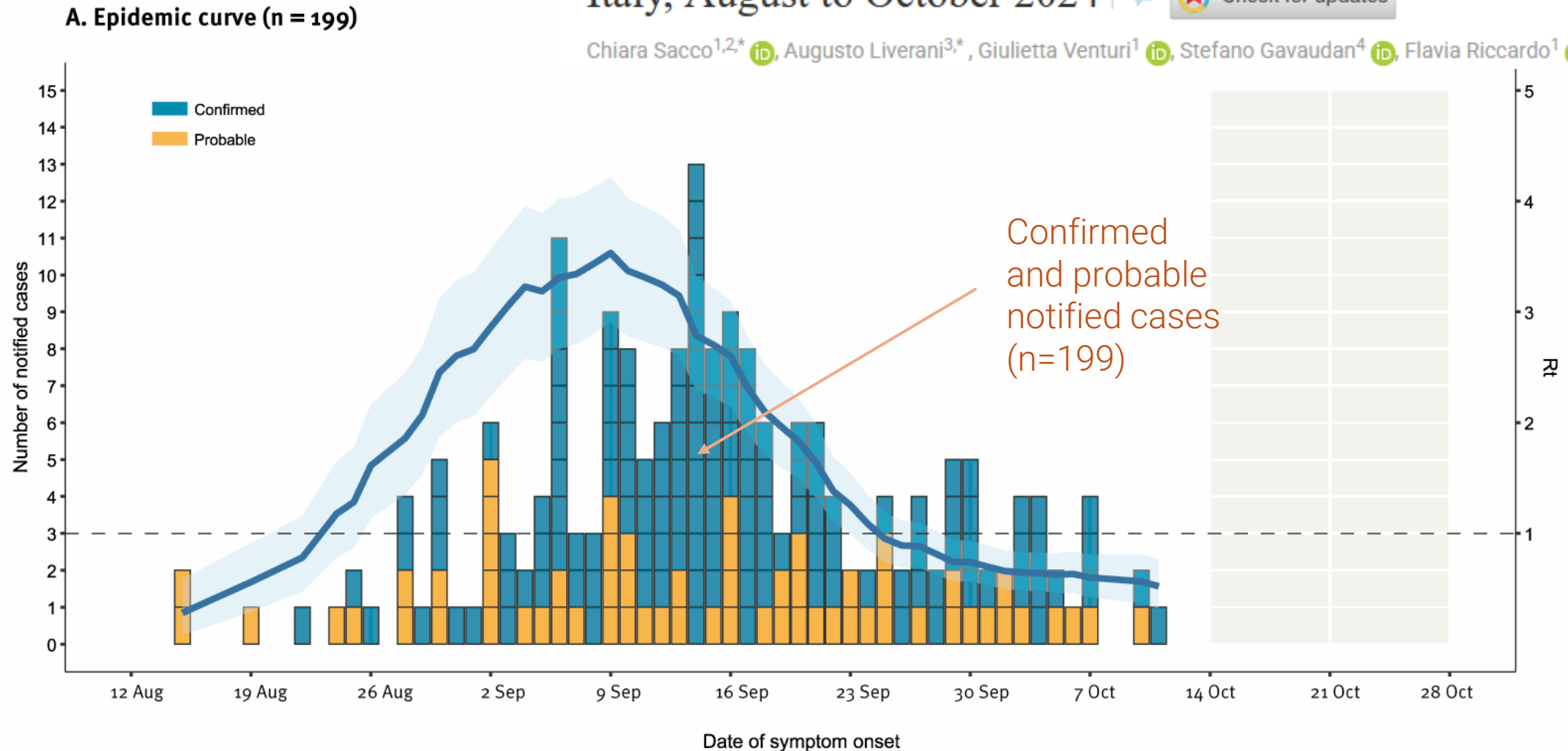
The emergence of dengue in Italy

Rapid communication

Autochthonous dengue outbreak in Marche Region, Central Italy, August to October 2024

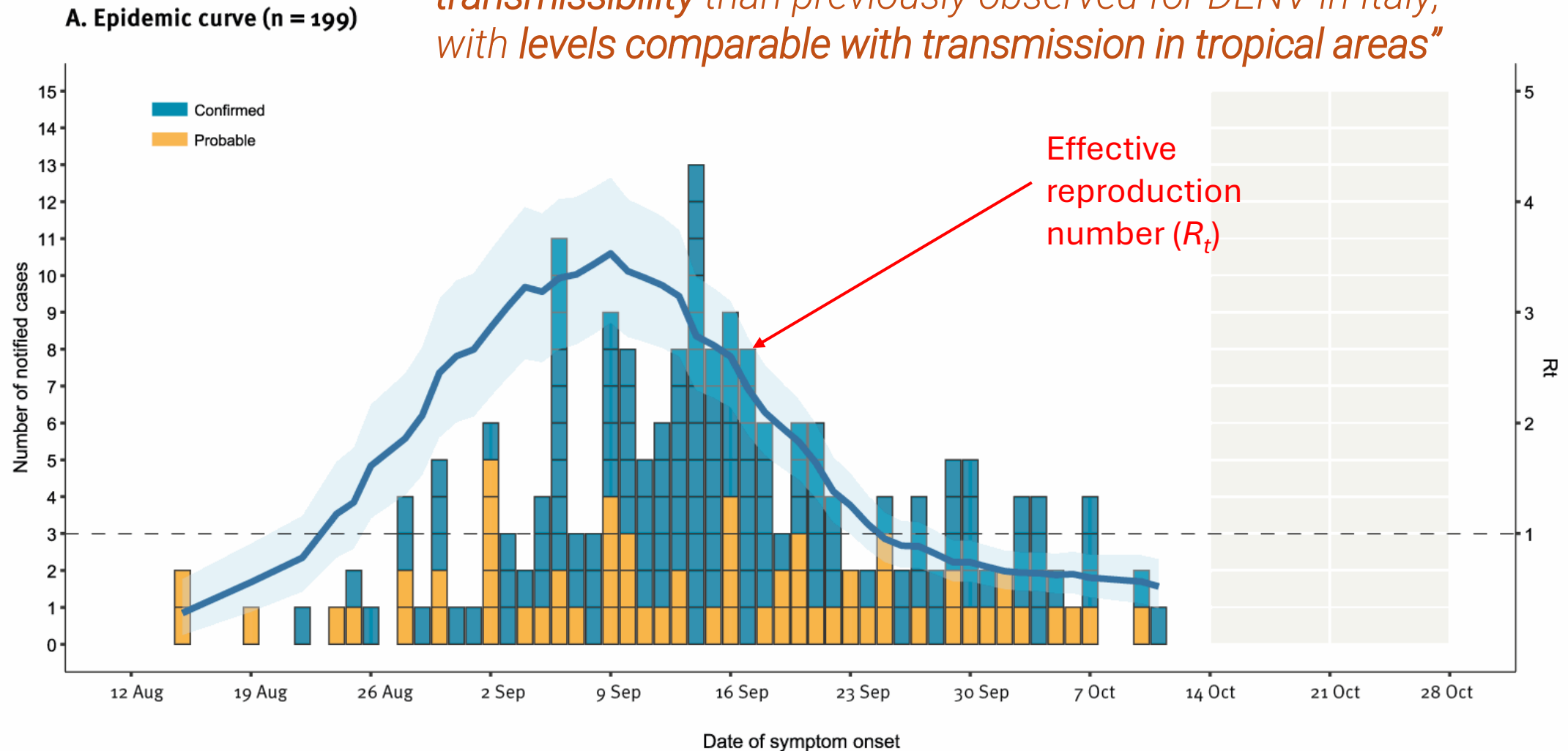


Chiara Sacco^{1,2,*} , Augusto Liverani^{3,*}, Giulietta Venturi¹ , Stefano Gavaudan⁴ , Flavia Riccardo¹ , Giov



The emergence of dengue in Italy

*“[This outbreak] marks a **notable shift** in the epidemiology of dengue in Europe. Our estimates suggest a **higher transmissibility** than previously observed for DENV in Italy, with **levels comparable with transmission in tropical areas**”*



The problem

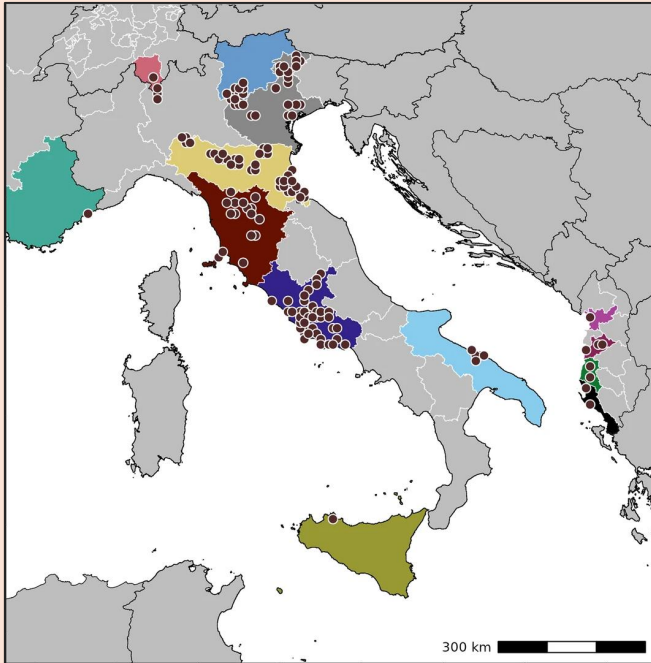
Mosquito borne diseases are now emerging in countries and regions that have not experienced comparable health threats in many decades.

The questions facing many decisionmakers include:

- Where and when will these diseases emerge?
- What data do we need to monitor and predict outbreak risks?
- What are the key factors driving these outbreaks?
- How can we reduce the risks that they will establish?
- How can we improve our preparedness before they arrive?

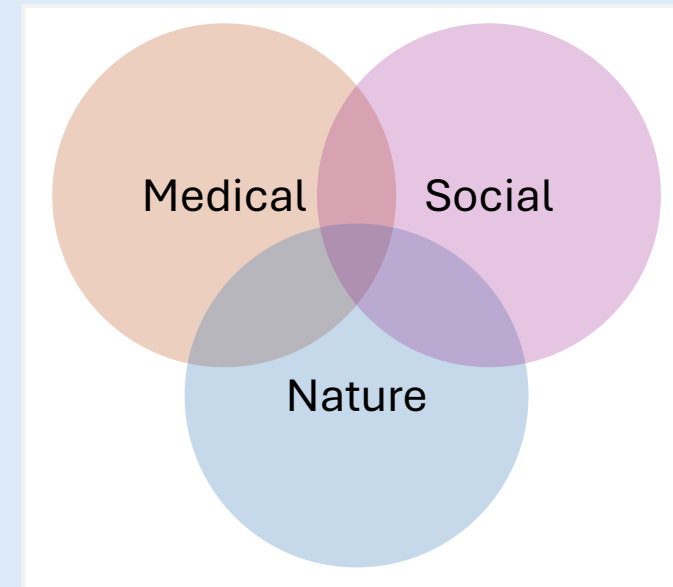
Today's two-part activity – in groups of 4-5 people

Mosquito data mapping (~50 mins)



RStudio: exploring and plotting
mosquito monitoring data

Group policy discussion (~50 mins)



Group discussion: brainstorm
solutions for dengue emergence

Mosquito data mapping

Head to this link and follow the instructions

https://github.com/rorygibb/pnl_summer_school

You'll need RStudio with the packages “sf”, “terra”, “dplyr” and “magrittr” installed.

Feel free to work on this groups – the goal is to explore and discuss the data, rather than detailed technical learning.

Group discussion: solutions to reduce outbreak risks



What types of intervention are possible?



Healthcare/health systems-based

(e.g. availability/resourcing of health centres, treatments, diagnostics, vaccines)



Behavioural

(e.g. population awareness, bite avoidance, sanitation/hygiene, workplace measures)



Infrastructure and design

(e.g. built environment design, water/sanitation infrastructure, housing)



Nature/ecosystem-based

(e.g. habitat connectivity, restoration, green-space design)

Nature-based (ecosystem-based) interventions

Protection of
existing habitat



Ecological
restoration or
creation of new
habitats



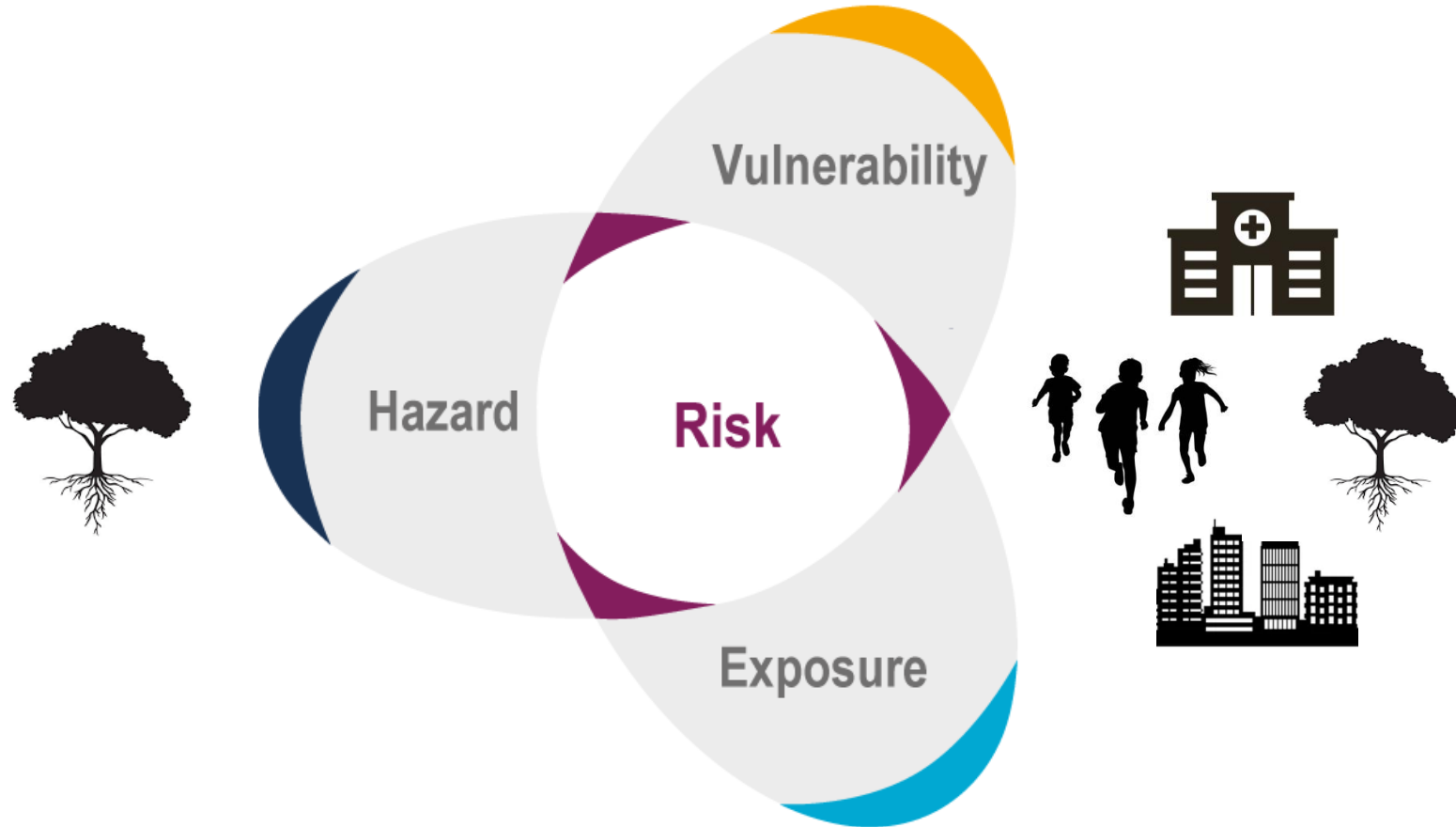
Improved
management of
natural/semi-
natural habitat



Improving
connectivity



What types of intervention are possible?



Group discussion: solutions to reduce outbreak risks

You are scientists advising policymakers after this outbreak, and they want to know...

- What are the factors that could be driving dengue transmission in Italy?
- What measures could support better response to future outbreaks?
- What solutions can be developed to reduce the risk of future outbreaks occurring (including nature-based solutions)?

In groups of 4-5, discuss and map drivers and potential solutions (~30mins). Each group will report back their ideas (1-2 mins each), and we will discuss and feed back.

Types of intervention



Timescales of intervention



What data or evidence is needed to enable decisions?